

CALCULUS II

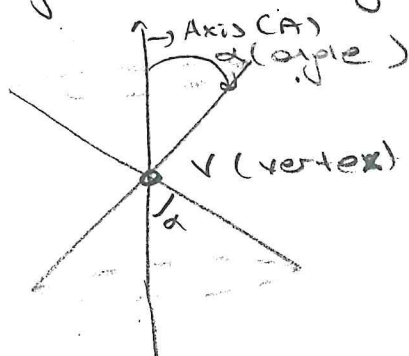
WEEK I

CONICS, PARAMETRIC CURVES AND POLAR CURVES

→ Here we give geometric definitions and standard definitions of circles, parabolas, ellipses and hyperbolas. They are called conic sections or conics since they result from intersecting a double cone , with a plane .

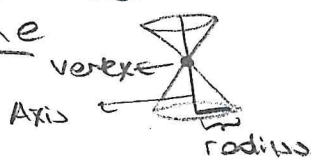
→ To be specific, suppose that a line A is fixed in space and V is a point on A .

→ The right-circular cone having axis (A) , ^(direction) vertex (V) and semi-vertical angle (α) is the surface consisting of all points on straight lines through V that make angle α with line A .



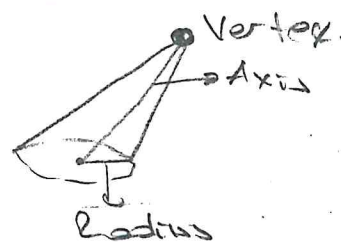
Types of cones

Right circular cone



Cross section with horizontal plane is circle 

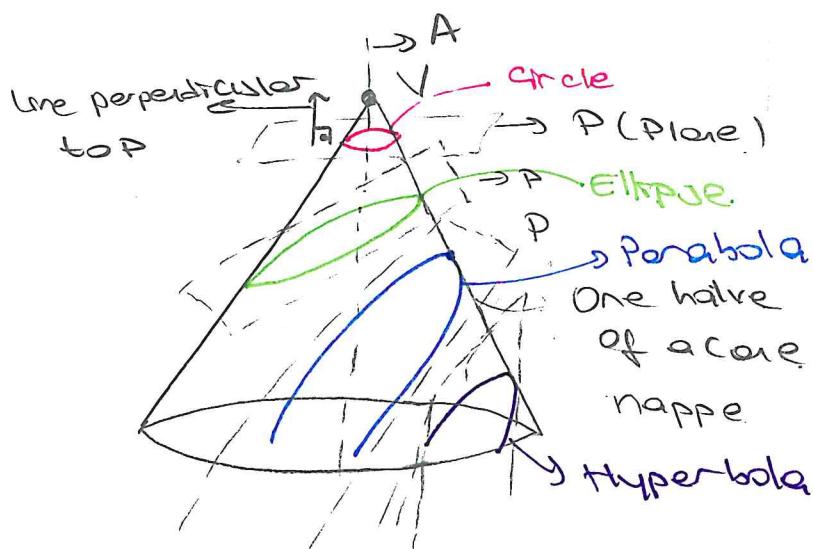
Oblique cone



Cross section with horizontal plane is Oval

(Oval: similar to ellipse but not ellipse 😊)

→ The cone has two halves (each called a nappe) lying on opposite sides of a vertex V . Any plane P that doesn't pass through V will intersect the cone in curve C .



→ If a line normal (perpendicular to P) makes θ angle with the axis A of the cone

$$0 \leq \theta \leq \pi/2$$

Circle if $\theta = 0$

Ellipse if $0 < \theta < \pi/2 - \alpha$

Parabola if $\theta = \pi/2 - \alpha$

Hyperbola if $\pi/2 - \alpha < \theta \leq \pi/2$

→ All conics can be represented analytically by a second-degree equation of the general form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

where $A, B, \dots, F$ are constants.

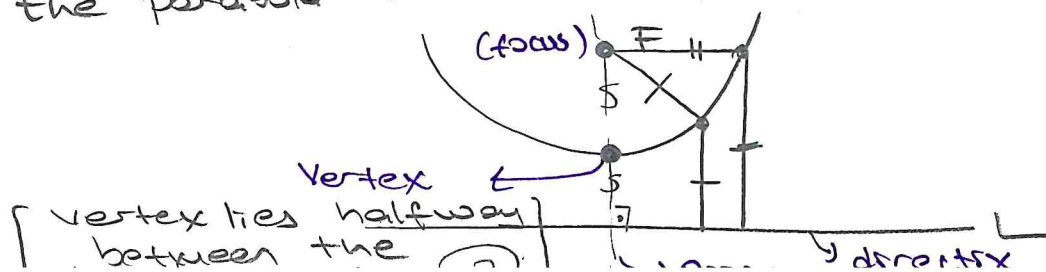
However, such an equation can also represent the empty set, a single point, or one or two straight lines if left-hand side factors into linear factors

$$(A_1x + B_1y + C_1)(A_2x + B_2y + C_2) = 0$$

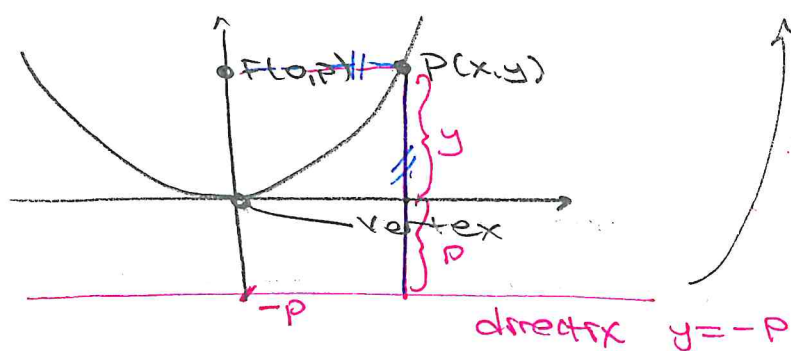
Now each is a line.

PARABOLA

A parabola consists of points in the plane that are equidistant from a given point (focus) and a given straight line (directrix). The line through the focus perpendicular to the directrix is called the principal axis (or axis) of the parabola. The vertex of the parabola is the point where the parabola crosses its principal axis.



Ex Consider the parabola whose vertex is at the origin $(0,0)$. The focus is $F = F(0,p)$.



Using the information
Vertex is in the half
of directrix and focus
we can say that
directrix is $y = -p$
 $|p|$: focal length
(related with
focus)

- Distance from the focus $F(0,p)$ to the point $P(x,y)$ on the parabola

$$|PF| = \sqrt{(x-0)^2 + (y-p)^2} = \sqrt{x^2 + (y-p)^2}$$

- Distance of $P(x,y)$ to the directrix is $|y+p|$. By definition of the parabola we have

$$\sqrt{x^2 + (y-p)^2} = |y+p|$$

$$x^2 + (y-p)^2 = (y+p)^2 \Rightarrow x^2 + y^2 - 2py + p^2 = y^2 + 2py + p^2$$

$$\boxed{x^2 = 4py}$$

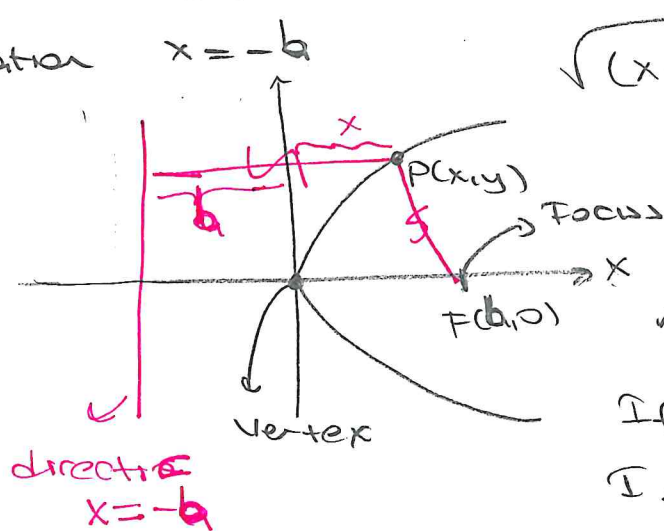
Equation of the parabola
with focus $(0,p)$ and
directrix $y = -p$

→ If we write $a = \frac{1}{4p}$ then the standard equation of the above parabola becomes $y = ax^2$

- The parabola opens upward if $p > 0$
- The parabola opens downward if $p < 0$

- Since $y = ax^2 = a(-x)^2$ it is symmetric w.r.t y axis

Ex 7/2 Find an equation of the parabola whose focus is the point $F=(b,0)$ and whose directrix is the line L with equation $x=-b$



$$\sqrt{(x-b)^2 + y^2} = |x+b|$$

$$(x-b)^2 + y^2 = (x+b)^2$$

$$x^2 - 2bx + b^2 + y^2 = x^2 + 2bx + b^2$$

$$y^2 = 4bx$$

If $b > 0$ opens right C

If $b < 0$ opens left D

Standard Equations For parabolas

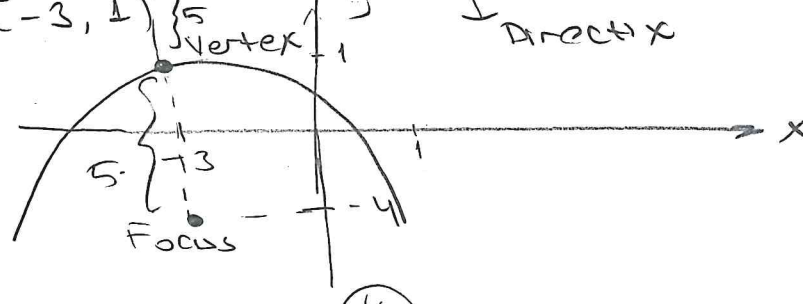
Focus	Vertex	Directrix	Equation
$(a,0)$	$(0,0)$	$x = -a$	$y^2 = 4ax$
$(-a,0)$	$(0,0)$	$x = a$	$y^2 = -4ax$
$(0,a)$	$(0,0)$	$y = -a$	$x^2 = 4ay$
$(0,-a)$	$(0,0)$	$y = a$	$x^2 = -4ay$
$(h,k+p)$	(h,k)	$y = k-p$	$(x-h)^2 = 4p(y-k)$
$(h+p,k)$	(h,k)	$x = h-p$	$(y-k)^2 = 4p(x-h)$

Ex 7/3 State the vertex and the focus of the following parabolas

(1) $(x+3)^2 = -20(y-1)$ $h = -3$ $k = 1$ $p = -5$
 $-20 = 4p$

Focus $(h, k+p) = (-3, 1-5) = (-3, -4)$

Directrix $y = 1 + 5 = 6$ focal length $|p| = 5$
 Vertex $(-3, 1)$

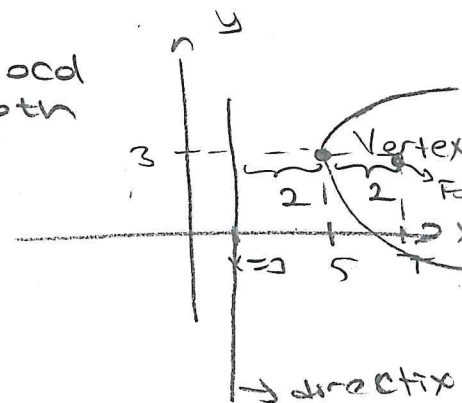


② $(y-3)^2 = 8(x-5)$

Vertex $(5, 3)$ $4p = 8$ $p = 2$ $|p| = 2$ focal length

Focus $(5+2, 3) = (7, 3)$

Directrix $x = 5 - 2 = 3$

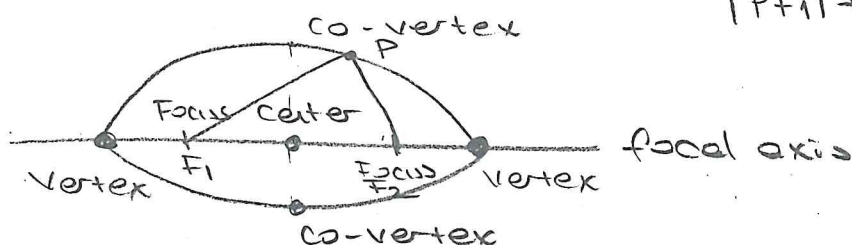


ELLIPSE

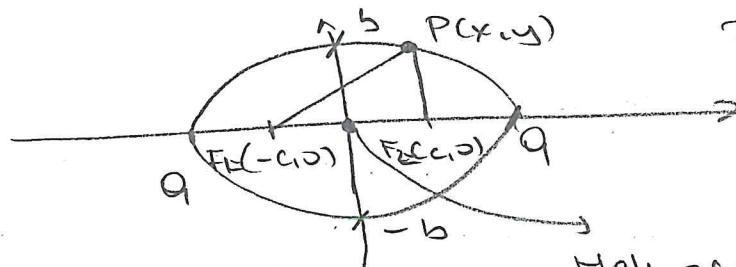
Set of points, in a plane whose distances from two fixed points in the plane have a constant sum

The two fixed points are the foci of the ellipse
(consider a rope (i.e.) someone hold from foci and one from rope and moving)

$|PF_1| + |PF_2|$ is CONSTANT



Ex 7 Let us consider an ellipse centered at the origin



If the foci are $F_1(-c,0)$, $F_2(c,0)$
and $|PF_1| + |PF_2|$ is denoted by $2a$,
then find equation
Half of the foci

$$|PF_1| + |PF_2| = 2a$$

$$\sqrt{(x+c)^2 + y^2} + \sqrt{(x-c)^2 + y^2} = 2a$$

$$\sqrt{(x+c)^2 + y^2} = 2a - \sqrt{(x-c)^2 + y^2}$$

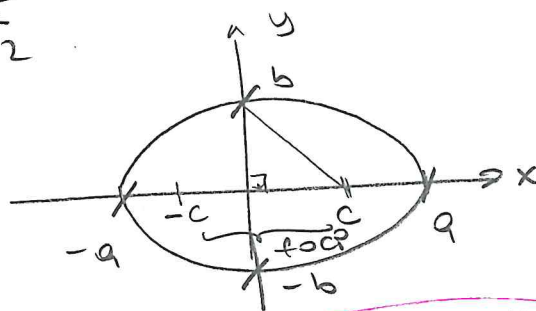
$$\underbrace{(x+c)^2 + y^2}_{x^2 + 2cx + c^2} = 4a^2 - 4a\sqrt{(x-c)^2 + y^2} + \underbrace{(x-c)^2 + y^2}_{x^2 - 2cx + c^2}$$

$$4a\sqrt{(x-c)^2 + y^2} = 4(a^2 - cx) \quad \sqrt{(x-c)^2 + y^2} = a - \frac{cx}{a}$$

$$x^2 - 2cx + c^2 + y^2 = a^2 - 2cx + \frac{c^2 x^2}{a^2}$$

$$y^2 + x^2 \left(1 - \frac{c^2}{a^2}\right) = a^2 - c^2$$

$$\frac{y^2}{a^2 - c^2} + \frac{x^2}{a^2} = 1$$



→ Let $b = \sqrt{a^2 - c^2}$, then we have

$$a \geq b > 0$$

$$b^2 = a^2 - c^2$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad a \geq b > 0$$

The following holds

a is semi-major

b is semi-minor

$c = \sqrt{a^2 - b^2}$ semi-focal

Foci $(\pm c, 0)$ where $c^2 = a^2 - b^2$ vertices $(\pm a, 0)$

Note If the foci of ellipse are located on the y axis interchange x and y in (A)

$$\frac{x^2}{a^2 - c^2} + \frac{y^2}{b^2} = 1$$

$$c^2 = a^2 - b^2 \quad a \geq b$$

Foci $(0, \pm c)$ $c^2 = a^2 - b^2$ vertices $(0, \pm a)$

Major axis

of the ellipse is the line segment of length $2a$ joining the lines $(a, 0)$ $(-a, 0)$

Minor axis

of the ellipse is the line segment $2b$ joining the points $(0, b)$ $(0, -b)$

Center to foci distance

$$c = \sqrt{a^2 - b^2} \quad (\text{Half way between the foci})$$

Semi major axis

The number a

Semi minor axis

The number b

Ex Consider the ellipse $25x^2 + 4y^2 + 100x - 40y + 100 = 0$

$$25(x+2)^2 + 4(y-5)^2 - 100 = 0$$

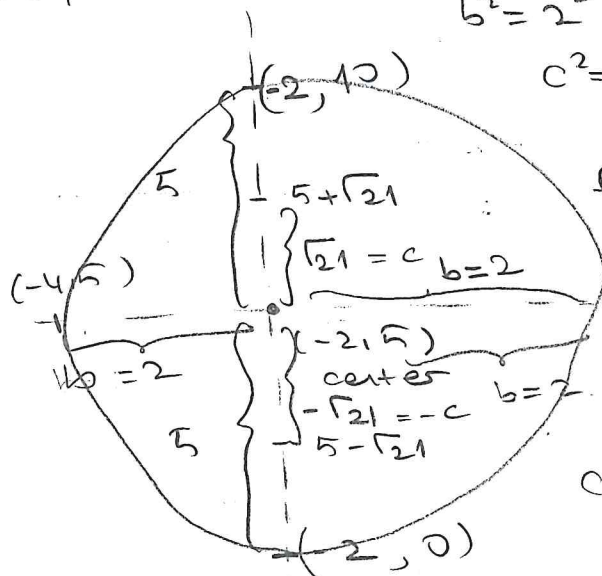
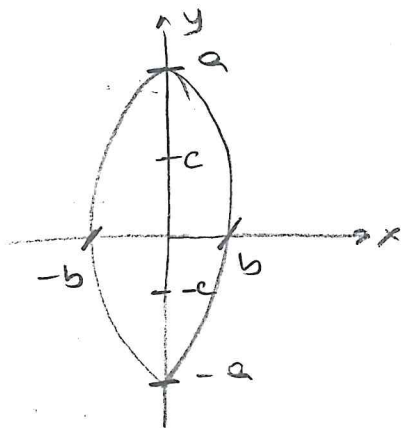
$\underbrace{\hspace{2cm}}_{x^2+4x+4} \quad \underbrace{\hspace{2cm}}_{y^2-10y+25}$

$$\frac{(x+2)^2}{2^2} + \frac{(y-5)^2}{5^2} = 1 \quad \text{Center } (-2, 5)$$

— $5 > 2$ lies vertically

→ The larger denominator is a^2 and the y part of the equation has the larger denominator, so this ellipse will be

taller than wider. So, we take $a^2 = 5^2 \Rightarrow a = 5$
 $b^2 = 2^2 \Rightarrow b = 2$



$$c^2 = a^2 - b^2 = 21$$

$$c = \pm\sqrt{21}$$

$$\text{foci } (-2, 5 + \sqrt{21})$$

$$(-2, 5 - \sqrt{21})$$

$$\text{Vertices } (-2, 10)$$

$$(-2, 0)$$

$$\text{Co-vertices } (0, 5)$$

$$(-4, 5)$$

Major axis length: 10
 Minor axis length: 4

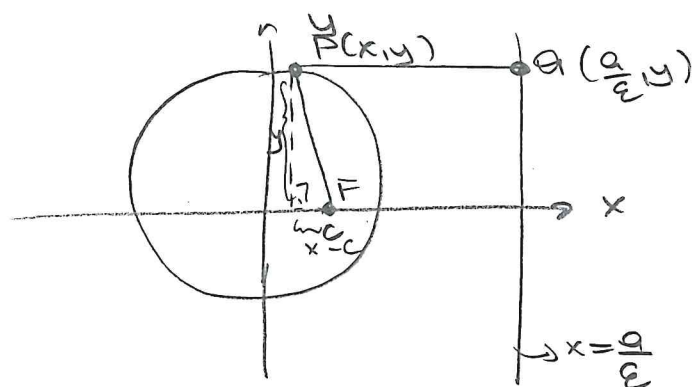
Eccentricity The eccentricity of an ellipse is the ratio of the semi-focal separator to the semi-major axis. We denote eccentricity ϵ . For the

ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $\epsilon = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a}$
 $a > b$

Note that $\epsilon < 1$ for any ellipse; the greater value of ϵ , the more elongated (less circular) is the ellipse. If $\epsilon = 0$ so that $a = b$ and $c = 0$, the two foci coincide and the ellipse is a circle

The Directrices of an Ellipse

→ If $a > b > 0$ each of the lines $x = a/\epsilon$ and $x = -a/\epsilon$ is called a directrix of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



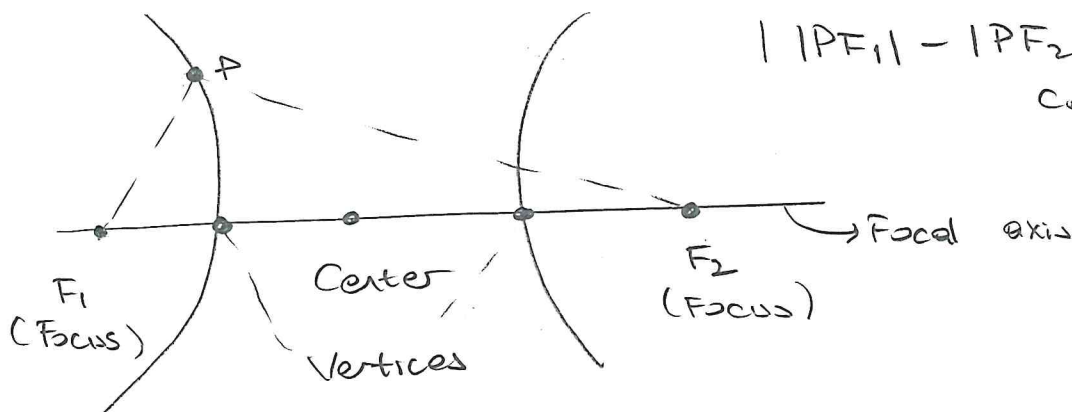
$$\begin{aligned}
 |PF|^2 &= (x-c)^2 + y^2 \\
 &= x^2 - 2cx + c^2 + b^2\left(1 - \frac{x^2}{a^2}\right) \\
 &= x^2\left(\frac{a^2 - b^2}{a^2}\right) - 2cx + a^2 - b^2 + b^2 \\
 \text{Use } c &= \epsilon a \\
 &= x^2 \frac{c^2}{a^2} - 2cx + a^2 \\
 &= \epsilon^2 x^2 - 2\epsilon ax + a^2 \\
 &= (a - \epsilon x)^2
 \end{aligned}$$

Thus, $PF = a - \epsilon x$. Also, $QP = a/\epsilon - x = \frac{(a - \epsilon x)}{\epsilon}$. Therefore, $\frac{PF}{QP} = \epsilon$

HYPERBOLAS

→ Set of points in a plane whose distances from two fixed points in the plane have a constant difference

→ The definition of hyperbola is similar to that of an ellipse the only difference is that the sum of distance has become a difference of distances.



$| |PF_1| - |PF_2| |$ is a constant!!

Ex 1 If the foci of a hyperbola are $F_1 = (c, 0)$ and $F_2 = (-c, 0)$ the difference of the distance from a point $P = (x, y)$ on the hyperbola to these foci is $2a$ (where $a < c$), then obtain the equation of the hyperbola

$$|PF_1| = \sqrt{(x-c)^2 + y^2}$$

$$|PF_2| = \sqrt{(x+c)^2 + y^2}$$

$$||PF_1| - |PF_2|| = 2a.$$

$$|PF_1| - |PF_2| = \begin{cases} 2a & \text{(right branch)} \\ -2a & \text{(left branch)} \end{cases}$$

$$\sqrt{(x-c)^2 + y^2} - \sqrt{(x+c)^2 + y^2} = 2a.$$

$$\left(\sqrt{(x-c)^2 + y^2} \right)^2 = \left(2a + \sqrt{(x+c)^2 + y^2} \right)^2$$

$$\underbrace{(x-c)^2 + y^2}_{x^2 - 2cx + c^2} = 4a^2 + 4a\sqrt{(x+c)^2 + y^2} + \underbrace{(x+c)^2 + y^2}_{x^2 + 2cx + c^2}$$

$$-4cx = 4a^2 + 4a\sqrt{(x+c)^2 + y^2}$$

$$-cx = a^2 + a\sqrt{(x+c)^2 + y^2}$$

$$\left(-a\sqrt{(x+c)^2 + y^2} \right)^2 = (a^2 + cx)^2$$

$$a^2((x+c)^2 + y^2) = a^4 + 2a^2cx + c^2x^2$$

$$a^2(x^2 + 2cx + c^2 + y^2) = a^4 + 2a^2cx + c^2x^2$$

$$a^2x^2 + 2a^2cx + a^2c^2 + a^2y^2 = a^4 + 2a^2cx + c^2x^2$$

$$(a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$$

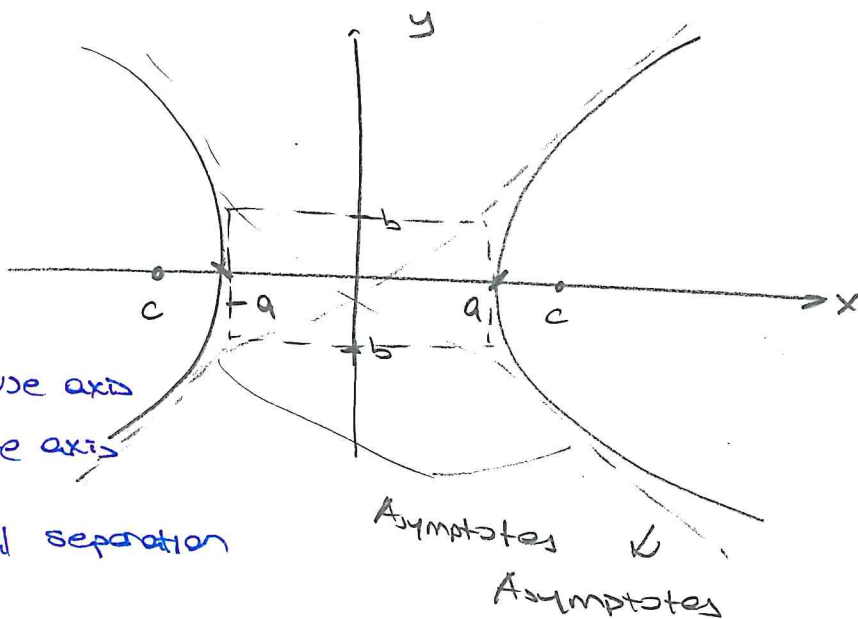
(a)

$$(a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$$

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1$$

(since $a < c$ $a^2 - c^2 < 0$)
let $a^2 - c^2 = -b^2$.

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}$$



a the semi-transverse axis

b the semi-conjugate axis

$c = \sqrt{a^2 + b^2}$ the semi-focal separation

Asymptotes How to obtain asymptotes? $y = \pm \frac{b}{a}x$

Solve $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$

$$y^2 = \frac{b^2}{a^2}x^2 \quad y = \pm \frac{b}{a}x \text{ asymptotes}$$

NOTE it's over formally in that way

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{solve it} \quad y^2 = \frac{b^2}{a^2}x^2 \left(1 - \frac{a^2}{x^2}\right)$$

$$y = \pm \frac{b}{a}x \sqrt{1 - \frac{a^2}{x^2}}$$

$x \rightarrow \infty \quad 1 - \frac{a^2}{x^2} \rightarrow 1$
and $y \rightarrow \pm \frac{b}{a}x$ is dominant

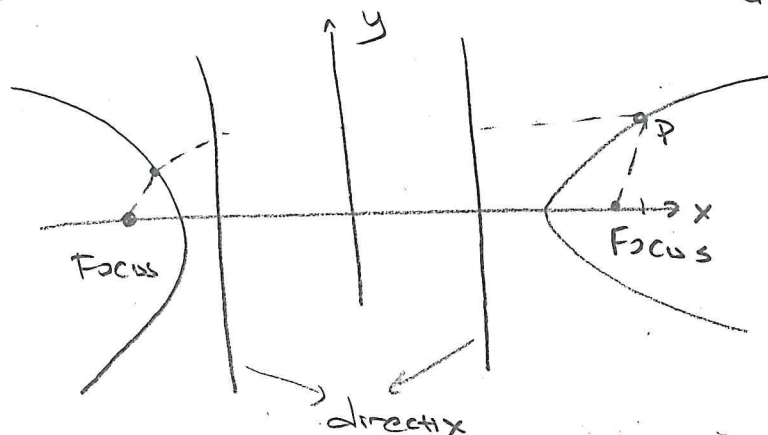
The eccentricity of the hyperbola

→ The eccentricity of the hyperbola is (eccentric)

$$e = \frac{c}{a} = \frac{\sqrt{a^2 + b^2}}{a}$$

Note that $e > 1$. The lines $x = \pm(a/e)$ are called the

directrices of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$



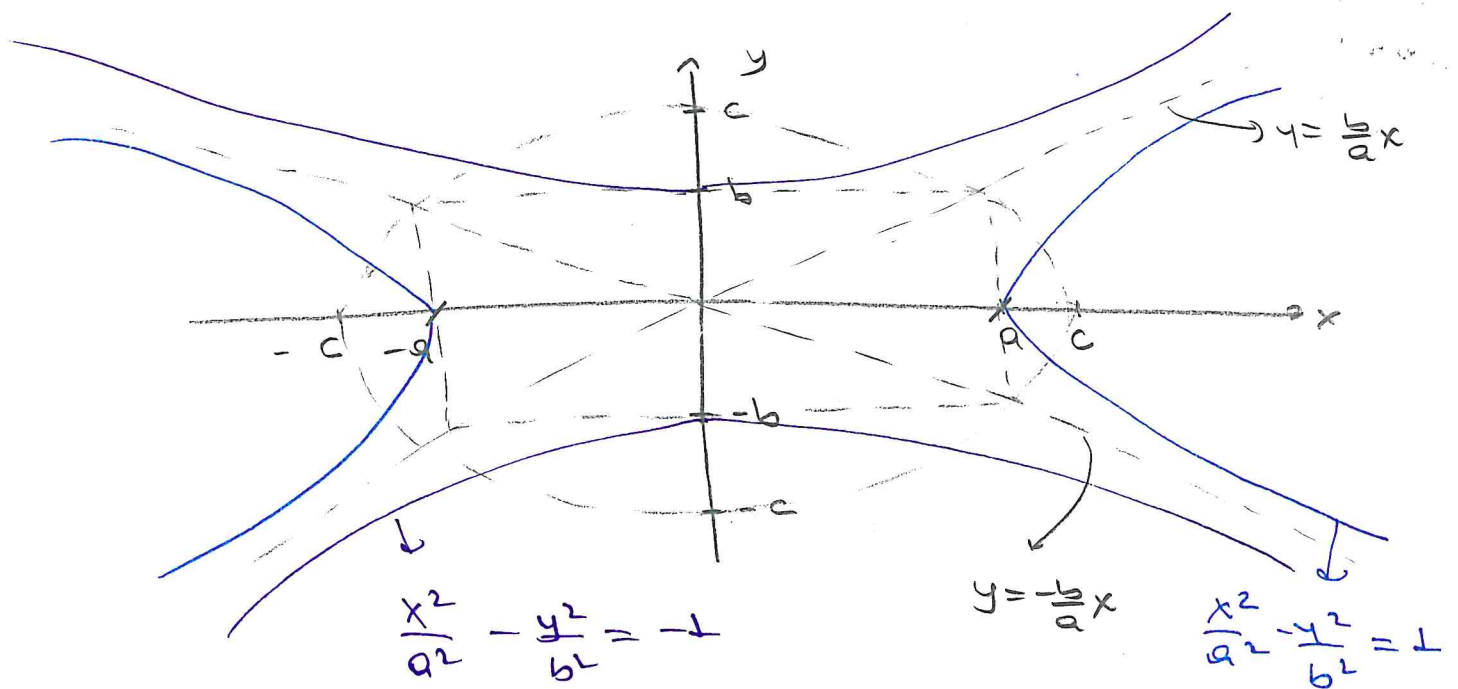
→
$$e = \frac{\text{distance from } P \text{ to focus}}{\text{distance from } P \text{ to the corresponding directrix}}$$

↓
eccentricity

→ A hyperbola with the same asymptotes as $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, but with transverse axis along the y-axis, vertices at $(0, b)$ and $(0, -b)$ and foci at $(0, c)$ and $(0, -c)$ is represented by the equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1 \text{ or, equivalently } \frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$$

→ These two hyperbolas are said to be conjugate to one another



Hyperbolas for more general form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Asymptotes $y-k = \pm \frac{b}{a}(x-h)$
Center (h, k)

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

Asymptotes $y-k = \pm \frac{a}{b}(x-h)$
Center (h, k)

Ex 7 Find the center, vertices and asymptotes of the equation

$$4x^2 - 5y^2 + 40x - 30y - 45 = 0$$

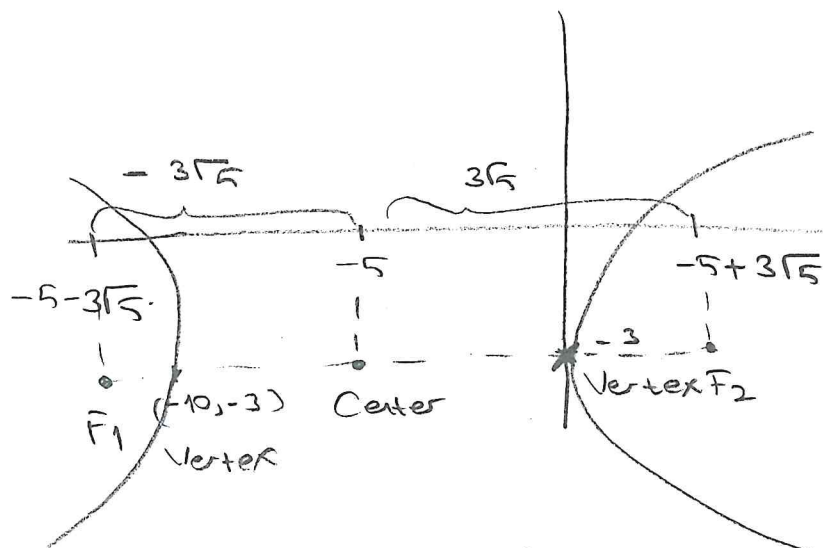
$$4(x^2 + 10x) - 5(y^2 + 6y) - 45 = 0$$

$$4(x^2 + 10x + 25) - 100 - 5(y^2 + 6y + 9) + 45 - 45 = 0$$

$$4(x+5)^2 - 5(y+3)^2 = 100$$

$$\frac{(x+5)^2}{25} - \frac{(y+3)^2}{20} = 1$$

$$a^2 = 25 \quad b^2 = 20 \quad c^2 = 25 + 20 = 45 \Rightarrow c = 73\sqrt{5}$$



Center $(-5, -3)$

Vertices $(-5-5, -3) \quad (-5+5, -3)$ $\Rightarrow$ What happens when $y = -3$
 $(-10, -3) \quad (0, -3)$ $\frac{(x+5)^2}{25} = 1 \quad x = -10 \quad x = 0$

Foci $(-5-3\sqrt{5}, -3)$
 $(-5+3\sqrt{5}, -3)$

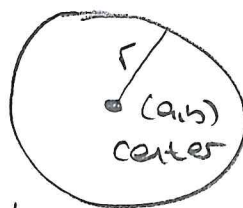
Asymptotes $y+3 = \pm \frac{2\sqrt{5}}{5}(x+5)$

Circle

Set of points in a plane whose distance from a given fixed point in the plane is constant

Circle of radius r centered

of the point (a, b) : $(x-a)^2 + (y-b)^2 = r^2$



CLASSIFYING GENERAL CONICS

→ Consider the quadratic equation in two variables

$$\boxed{A'x^2 + B'xy + C'y^2 + D'x + E'y + F' = 0 \quad (A')^2 + (B')^2 + (C')^2 \neq 0} \quad (*)$$

generally, represents a conic curve, but in certain degenerate cases it may represent two straight lines ($x^2 - y^2 = 0$ represents the lines $x = y$ and $x = -y$), one straight line ($x^2 = 0$ represent the line $x = 0$), a single point ($x^2 + y^2 = 0$ represents the origin), or no points at all ($x^2 + y^2 = -1$ is not satisfied by any point in the plane)

→ By rotating the coordinate axes, we can eliminate the term with the product x and y (The details will not be given) ^{1/11} $(*)$ (see summary page 2)

Then, we get

$$\boxed{Ax^2 + Cy^2 + Dx + Ey + F = 0} \quad (*)$$

- a circle If $A = C \neq 0$ (special cases: the graph is a point or there is no graph at all)
- a parabola If $(*)$ is quadratic in one variable (power 2) and linear (power 1) in the other
- an ellipse If A and C are both positive and both negative (special cases: circles, a point, no graph at all)
 $\text{such as } (x^2 + y^2 = 1) \quad (x^2 + y^2 = 0) \quad (x^2 + y^2 = -1)$
- an hyperbola If A and C have opposite signs
 (special case: a pair of intersecting lines for ex. $x^2 - y^2 = 0 \quad x = \pm y$)
- a straight line If A and C are zero and at least one of D and E is different from zero.
- one or two straight lines If the left hand side of $(*)$ can be factored into product of two linear factors

THE DISCRIMINANT TEST

$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ (Be careful about special cases written above)

- (a) a parabola if $B^2 - 4AC = 0$
- (b) an ellipse if $(A \neq C \text{ or } B \neq 0) \quad B^2 - 4AC < 0$
- (c) a circle if $B^2 - 4AC < 0$ and $A = C$
- (d) a hyperbola if $B^2 - 4AC > 0$

Ex 1 $2x^2 - xy + 5y^2 + 4x + 7 = 0$ let's use discriminant test

$A = 2 \quad B = -1 \quad C = 5 \quad B^2 - 4AC = 1 - 4 \cdot 2 \cdot 5 = -39 < 0$

Discriminant test says it is an ellipse

Ex 2 $xy - 5y^2 - 7x + 5y + 1 = 0$

$A = 0 \quad B = 1 \quad C = -5 \quad D = -7 \quad E = 5 \quad F = 1$

$B^2 - 4AC = 1 - 4 \cdot 0 \cdot (-5) = 1$ hyperbola

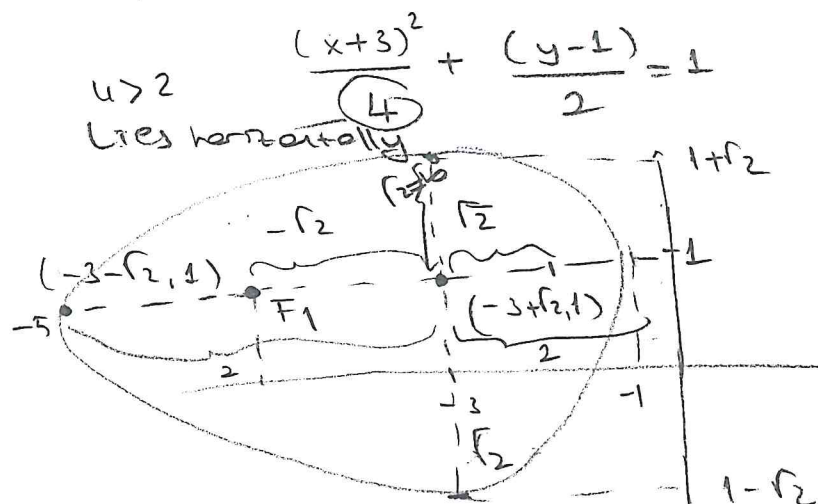
Ex 3 Describe the curve with equation $x^2 + 2y^2 + 6x - 4y + 7 = 0$

Complete squares in the x and y terms, and rewrite the equation as

$$x^2 + 6x + 2(y^2 - 2y) + 7 = 0$$

$$(x+3)^2 + 2(y^2 - 2y + 1) - 11 + 7 = 0$$

$$(x+3)^2 + 2(y-1)^2 = 4$$



Center $(-3, 1)$

$a = 2 \quad c = \sqrt{a^2 - b^2} = \sqrt{2}$
 $b = \sqrt{2}$

Foci $(-3 \pm \sqrt{2}, 1)$

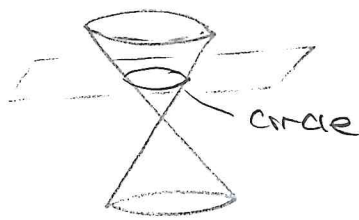
Vertices $(-1, 1) \quad (-5, 1)$

Co vertices $(-3, 1 + \sqrt{2})$

$(-3, 1 - \sqrt{2})$

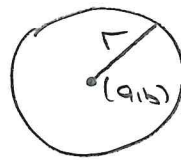
(15)

SUMMARY PAGE ①



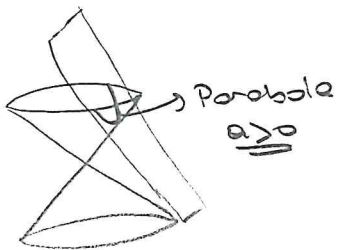
Circle

Circle $(x-a)^2 + (y-b)^2 = r^2$



r : radius

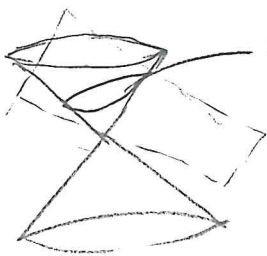
M : $M=(a,b)$ center



Parabola
 $a > 0$

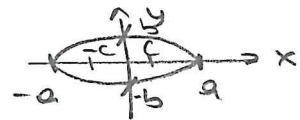
Parabola

Focus	Vertex	Directrix	Equation	Opens
$(a,0)$	$(0,0)$	$x = -a$	$y^2 = 4ax$	Up
$(-a,0)$	$(0,0)$	$x = a$	$y^2 = -4ax$	Down
$(0,a)$	$(0,0)$	$y = -a$	$x^2 = 4ay$	Right
$(0,-a)$	$(0,0)$	$y = a$	$x^2 = -4ay$	Left
$(h,k+p)$	(h,k)	$y = k-p$	$(x-h)^2 = 4p(y-k)$	
$(h+p,k)$	(h,k)	$x = h-p$	$(y-k)^2 = 4p(x-h)$	

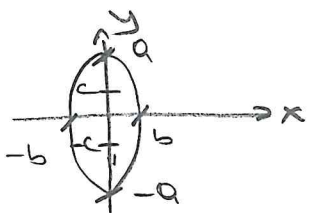


Ellipse

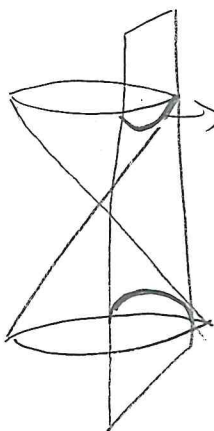
Focus on x axis $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$)
 center-to-focus distance $c = \sqrt{a^2 - b^2}$
 Focus $(\pm c, 0)$
 Vertices $(\pm a, 0)$



Focus on y axis $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ ($a > b$)
 center-to-focus $c = \sqrt{a^2 - b^2}$
 Focus $(0, \pm c)$ Vertices $(0, \pm a)$

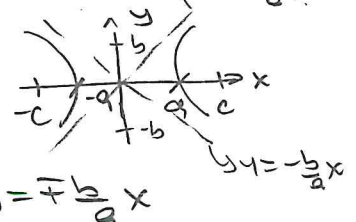


In each case a is the semi-major $-b$
 b is the semi-minor

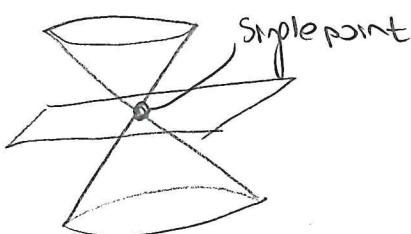


Hyperbola

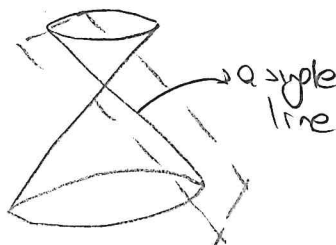
Hyperbolas
Focus on x-axis $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
 center-to-focus distance $c = \sqrt{a^2 + b^2}$
 Vertices $(\pm a, 0)$ Asymptotes $y = \pm \frac{b}{a}x$



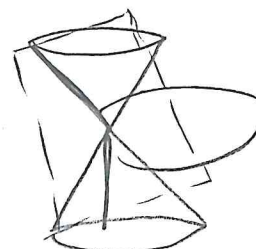
Focus on y-axis $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$
 center-to-focus distance $c = \sqrt{a^2 + b^2}$
 Focus $(0, \pm c)$ Vertices $(0, \pm a)$ Asymptotes $y = \pm \frac{a}{b}x$



Single point



Circle



pair of intersecting lines

SUMMARY PAGE 2

Classifying conic sections

① We have the form $Ax^2 + Cy^2 + Dx + Ey + F = 0$ (*)

- circle $A = C \neq 0$

- ellipse $A, C > 0$ or $A, C < 0$

- parabola A or C equals to zero

- hyperbola $A \cdot C < 0$

- straight line $A = C = 0$

- two straight lines write (*) as a product of two lines

② If we have the form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$

Discriminant test Circle $B^2 - 4AC < 0$ $A = C$

Ellipse $B^2 - 4AC < 0$ $A \neq 0$ or $B \neq 0$

Hyperbola $B^2 - 4AC > 0$

Parabola $B^2 - 4AC = 0$

Rotating (Not Necessary for students)

Ex Identify the curve $xy = 1$

Since $B^2 - 4AC > 0$ it is a hyperbola. But how?

We will rotate $x-y$ coordinate to $u-v$ coordinate by rotating xy axes with angle θ , then we get

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

$$A'u^2 + B'uv + C'v^2 + D'u + E'v + F = 0$$

$$\text{such that } A' = \frac{1}{2} (A(1 + \cos 2\theta) + B \sin 2\theta + C(1 - \cos 2\theta))$$

$$B' = (C - A) \sin 2\theta + B \cos 2\theta$$

$$C' = \frac{1}{2} (A(1 - \cos 2\theta) - B \sin 2\theta + C(1 + \cos 2\theta))$$

$$D' = D \cos \theta + E \sin \theta \quad \text{Remains unchanged}$$

$$E' = -D \sin \theta + E \cos \theta$$

If we choose θ $\tan 2\theta = \frac{B}{A - C}$ or $\theta = \pi/4$ if $A = C, B \neq 0$

then $B' = 0$, and the new equation doesn't have uv terms

Based on this formula rotating xy axis with $\theta = \pi/4$, we get

$$x = \frac{1}{\sqrt{2}} (u - v) \quad y = \frac{1}{\sqrt{2}} (u + v) \quad \text{so } xy = 1 \text{ gives}$$

$$u^2 - v^2 = 2 \text{ on } uv \text{ axis.}$$

PARAMETRIC CURVES

- Suppose we want to plot the path of a particle moving in a plane. This path looks like a curve, but we can't plot it like a curve in the Cartesian plane the reason for this is the fact that we can't express y directly in terms of x or x in terms of y .

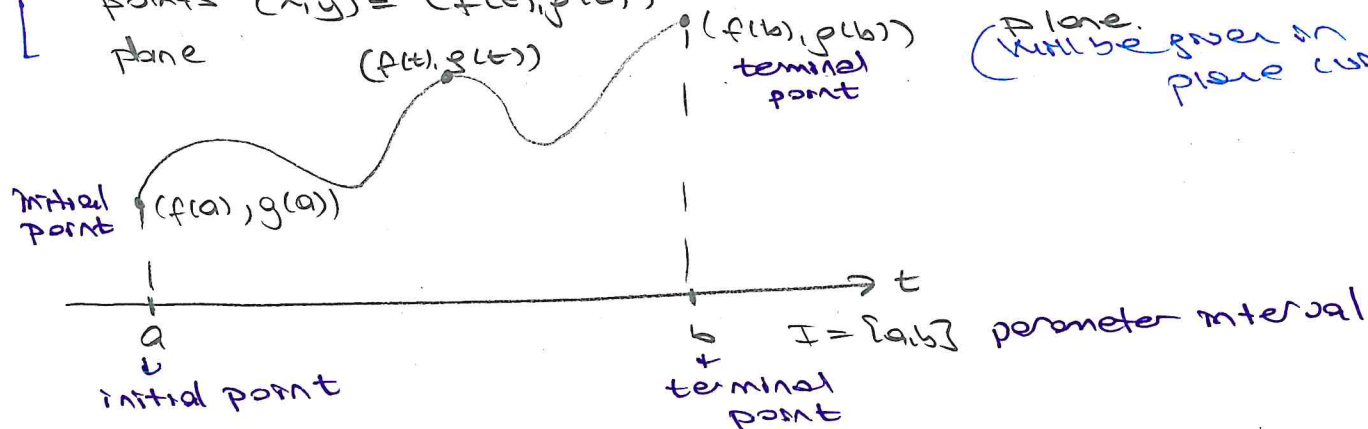
- To get around this problem, we can describe the path of the particle with a pair of equations $x=f(t)$ and $y=g(t)$.

Definition Parametric Curves A parametric curve C in the plane consists of an ordered pair (f,g) of continuous functions each defined on the same interval I . The equations

$$x=f(t), y=g(t), \text{ for } t \text{ in } I$$

are called parametric equations of the curve C . The independent variable t is called the parameter.

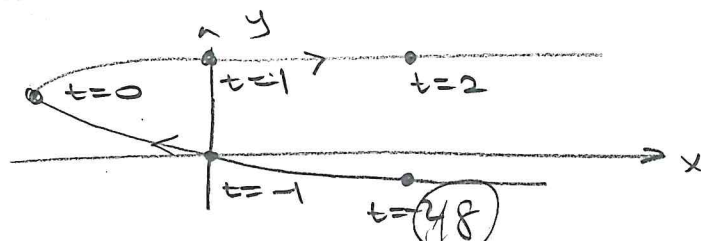
→ If x and y are given as continuous functions, $x=f(t)$, $y=g(t)$ over an interval of t -values, then the set of points $(x,y)=(f(t),g(t))$ is a curve in the coordinate plane. (Will be given in plane curves)



Ex 7 Sketch and identify the parametric curve

$$x=t^2-1 \quad y=t+1 \quad (-\infty < t < \infty)$$

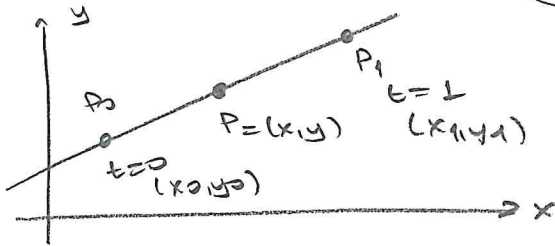
Parameter elimination $t = y-1 \quad x = (y-1)^2-1$



Ex (Parametric equations of a straight line) The straight line passes through two points $P_0 = (x_0, y_0)$ and $P_1 = (x_1, y_1)$ has parametric equations

$$x = x_0 + t(x_1 - x_0) \quad (-\infty < t < \infty)$$

$$y = y_0 + t(y_1 - y_0)$$



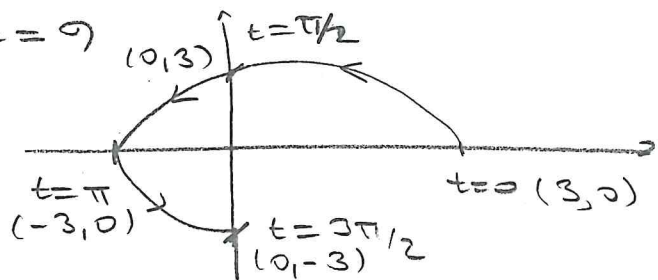
$$\frac{y - y_0}{x - x_0} = \frac{y_1 - y_0}{x_1 - x_0} = \text{constant}$$

(assuming $x_1 \neq x_0$)

$$0 \leq t \leq 1 \quad \begin{matrix} t=0 & (x_0, y_0) \\ t=1 & (x_1, y_1) \end{matrix}$$

Ex (An arc of a circle) Sketch and identify the curve $x = 3 \cos t$
 $y = 3 \sin t \quad (0 \leq t \leq 3\pi/2)$

$$x^2 + y^2 = 9 \cos^2 t + 9 \sin^2 t = 9$$



NOTE Parametric equation of a unit circle

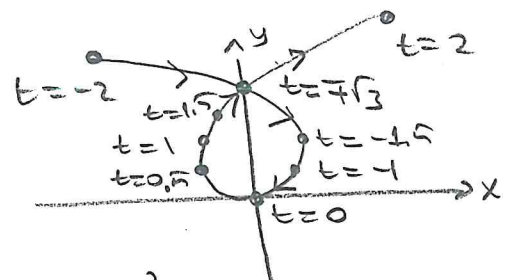
$$x = \cos t \quad y = \sin t \quad x^2 + y^2 = 1 \quad 0 \leq t \leq 2\pi$$

Ex (Parametric equation of an ellipse) Sketch and identify the curve $x = a \cos t$ $y = b \sin t$ $(0 \leq t \leq 2\pi)$ where $a > b > 0$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \cos^2 t + \sin^2 t = 1$$

Ex Sketch the parametric curve

$$x = t^3 - 3t \quad y = t^2 \quad (-2 \leq t \leq 2)$$



$$x = t(t^2 - 3) \Rightarrow x^2 = t^2(t^2 - 3)^2 = y(y - 3)^2 \Rightarrow \text{Symmetry wrt } y \text{ axis. Because } x \text{ is odd func. of } t \text{ and } y \text{ is even.}$$

t	-2	-3/2	-1	-1/2	0	1/2	1	3/2	2
x	-2	9/8	2	11/8	0	-11/8	-2	-9/8	2
y	4	9/4	1	1/4	0	1/4	1	9/4	4

Ex 7 Find the parametric equation for the right branch of the hyperbola $x^2 - y^2 = 1$.

Here we can't use $x = \cos t$, $y = \sin t$. But, we know that.

$$\sec^2 t = \tan^2 t + 1 \Rightarrow \sec^2 t - \tan^2 t = 1$$

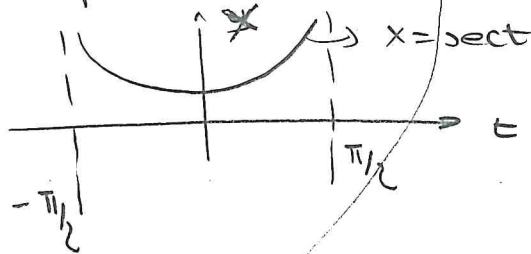
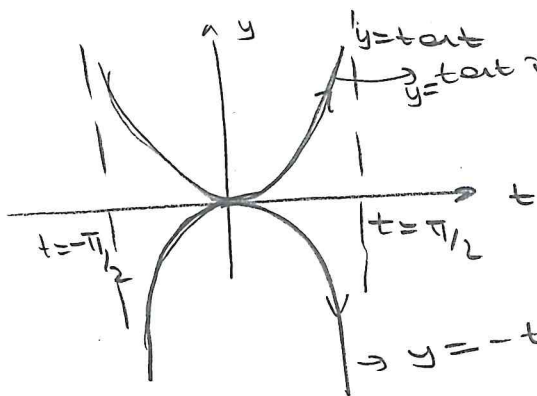
So let $x = \text{sect}$, $y = \text{tert}$

$-\pi/2 < t < \pi/2$

Right branch $x = \bar{7}$ sect since on right branch $x > 0$ we

have $x \Rightarrow \exists e \in t$

$$-\pi/2 < t < \pi/2$$



How do we choose the way of growth

$y = \mp \text{tert}$ dan be. choose $y = \text{tert} \uparrow \quad -\pi/2 < t < \pi/2$ $\swarrow$
 $y = -\text{tert} \downarrow \quad -\pi/2 < t < \pi/2$ $\searrow$

we choose $x = \text{sect}$

For example $t = -\pi/4$ $y = 1$
 $t = \pi/4$ $y = -1$
 $y = \tan t$

NOTE Remember Domain and Range of Trigonometric Function

<u>Function</u>	<u>Domain</u>	<u>Range</u>
$\sin$	$\mathbb{R}$	$[-1, 1]$
$\cos$	$\mathbb{R}$	$[-1, 1]$
$\tan$	$\mathbb{R} \setminus x = (2n+1)\pi/2, n \in \mathbb{Z}$	$\mathbb{R}$
$\cot$	$\mathbb{R} \setminus x = n\pi$	$\mathbb{R}$
cosec	$\mathbb{R} \setminus x = n\pi$	$\mathbb{R} \setminus (-1, 1)$
$\sec$	$\mathbb{R} \setminus x = (2n+1)\pi/2, n \in \mathbb{Z}$	$\mathbb{R} \setminus (-1, 1)$

→ In the right branch $x > 0$ so we choose $x = \sec t$ with

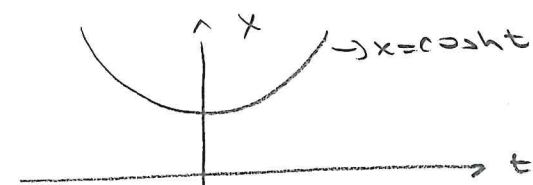
- $-\pi/2 < t < \pi/2$. If we choose $x = \text{tert}$ it will increase. $\Rightarrow$

$-\pi/2 < t < \pi/2$. So we have (20) If we choose $y = -\tan$
R+1 is fine

Also, we can choose $x = \cosh t$ $y = \sinh t$ $-\infty < t < \infty$

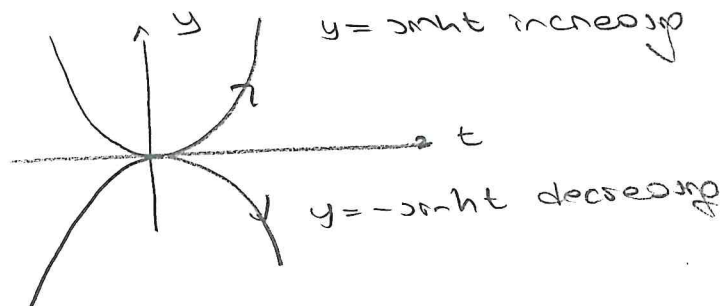
Here, $\cosh^2 t - \sinh^2 t = 1$

$x = \pm \cosh t$ $y = \pm \sinh t$



$$\cosh t = \frac{e^t + e^{-t}}{2}$$

$$\sinh t = \frac{e^t - e^{-t}}{2}$$

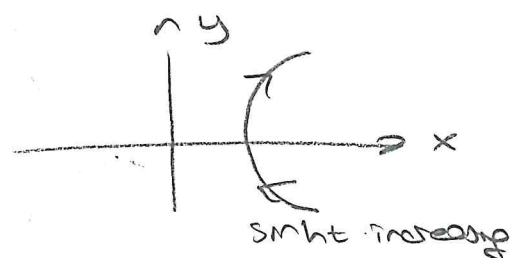


Since we have $x > 0$ on the right branch $x = \cosh t$

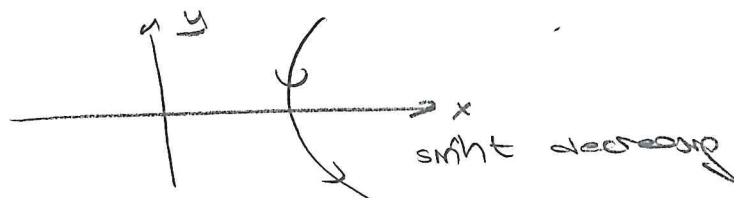
$y = \sinh t$ can be positive or negative $-\infty < t < \infty$

So for $(x, y) = (\cosh t, \sinh t)$

$-\infty < t < \infty$



$(x, y) = (\cosh t, \sinh t)$



Let, $x^2 - y^2 = 1$ $(x, y) = (a \sec t, b \tan t)$

Then for general case

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$x = a \sec t$
 $y = b \tan t$

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$x = a \sec t + h$
 $y = b \tan t + k$

→ Without trigonometric functions we can also use

$$x^2 - y^2 = 1$$

$\Rightarrow y = t$

$$x = \pm \sqrt{1+t^2}$$

Ex 7 Parametrize $x = y^2 - 4y + 3$. Let $y = t$ then $x = t^2 - 4t + 3$
 So, $(t^2 - 4t + 3, t)$ as t varies is a parametrization of the
 given parabola

Ex 7 Find Cartesian equation for $x = 3 - 3t$ $y = 2t$ $0 \leq t \leq 1$
 $y = 2t \Rightarrow t = y/2$ So $x = 3 - \frac{3y}{2}$ $2x + 3y = 6$ which
 is a straight line

General Plane Curves and Parametrization

Plane Curve

A plane curve is a set of points (x, y) in the plane such
 that $x = f(t)$ and $y = g(t)$ for some t in an interval I ,
 where f and g are continuous functions defined on I .
 Any such interval I and function pair (f, g) that generate
 points of C is called a parametrization of C .

(**) Since a plane curve doesn't involve any specific parametrization
It has no specific direction

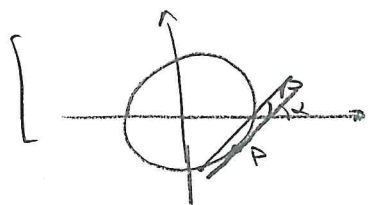
Ex 7 The circle $x^2 + y^2 = 1$ is a plane curve. Each of the
 following is a possible parametrization of the circle

- (1) $x = \cos t$, $y = \sin t$ ($0 \leq t \leq 2\pi$)
- (2) $x = \sin s^2$, $y = \cos s^2$ ($0 \leq s \leq \sqrt{2\pi}$)
- (3) $x = \cos(\pi(u+1))$, $y = \sin(\pi(u+1))$ ($-1 \leq u \leq 1$)
- (4) $x = 1 - t^2$, $y = t\sqrt{2 - t^2}$ ($-\sqrt{2} \leq t \leq \sqrt{2}$)

Ex 7 If f is a continuous function on an interval I , then the
 graph of f is a plane curve. One obvious parametrization
 of this curve is

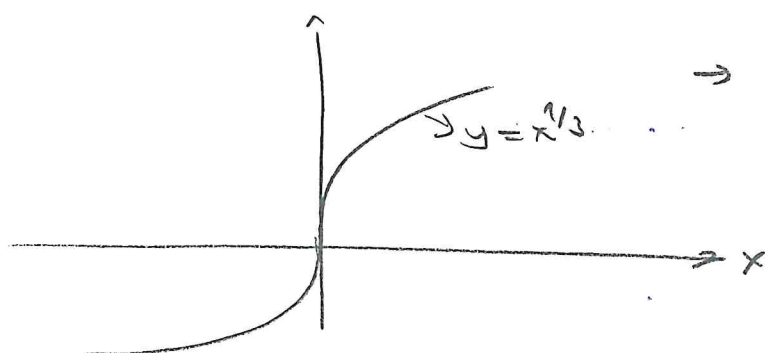
$$x = t, \quad y = f(t) \quad (t \in I)$$

→ We say that a plane curve is smooth if it has a tangent line at each point P and this tangent turns in a continuous way as P moves along the curve (That is, the angle between the tangent line and some fixed line, the x -axis say, is a continuous function of the position P)



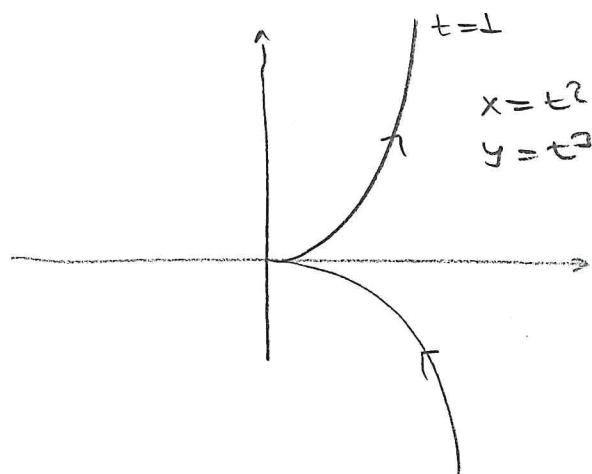
x and P is a correspondence of $P(x,y)$ x,y coordinates

A curve $y=f(x)$ is smooth on any interval where f' exists and continuous.



→ A curve may be smooth on an interval containing isolated singularities

(e.g. see the curve $y=x^{1/3}$
 $y' = \frac{1}{3} x^{-2/3}$)



This curve is not smooth at the origin but has a

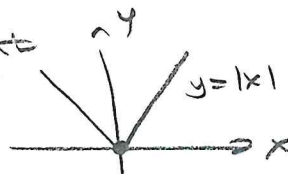
cusp (derivatives are zero, and directional derivative different sides has different signs)

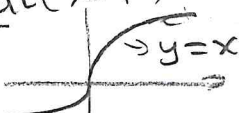
$x=t^2$ means $x=y^{2/3}$

$y=t^3$

Parametric case $x=f(t)=t^2$
 $y=g(t)=t^3$ $f'(0)=g'(0)=0$

$\lim_{y \rightarrow 0^+} x' = \infty$

Corner point 
 $(a, f(a))$
corner
 $\lim_{x \rightarrow a} f'(x)$ doesn't exist for $y=|x|$ $x=0$

Vertical Tangent (x_0, T)
 $y=x^{1/3}$
 $x=a \Rightarrow V.T.$ 

$\lim_{x \rightarrow a^+} f'(x) = \lim_{x \rightarrow a^-} f'(x) = \infty$
doesn't exist $\lim_{x \rightarrow a} -f'(x) = \lim_{x \rightarrow a^+} f'(x) = -\infty$

Cusp $y=x^{2/3}$ 

$(a, f(a))$
 $\lim_{x \rightarrow a^+} f'(x) = \infty$ and
 $\lim_{x \rightarrow a^-} f'(x) = -\infty$ for $y=x^{2/3}$ $x=0$
OR
 $\lim_{x \rightarrow a^+} f'(x) = -\infty$ and $\lim_{x \rightarrow a^-} f'(x) = \infty$
cusp

For $x=t^2, y=t^3$ Cartesian equation $y^2=x^3$ or $x=y^{2/3}$

$$\frac{dx}{dy} = \frac{2}{3} y^{-1/2} \quad \lim_{y \rightarrow 0^+} \frac{dx}{dy} = +\infty$$

NO TANGENT when $t=0$

THM The slope of a Parametric Curve

Let C be the parametric curve $x=f(t), y=g(t)$ where $f'(t)$ and $g'(t)$ are continuous on an interval I . If $f'(t) \neq 0$ on I , then C is smooth and has at each t a tangent line

$$\frac{dy}{dx} = \frac{g'(t)}{f'(t)}$$

Explanation is on page 34
Chain Rule

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \frac{dx}{dt} \neq 0$$

If $g'(t) \neq 0$ on I , then C is smooth and has at each t a normal line with slope

$$-\frac{dx}{dy} = -\frac{f'(t)}{g'(t)}$$

Thus, C is smooth except possibly at points where $f'(t)$ and $g'(t)$ or both zero

WARNING If f' and g' are continuous, and both vanish at some point t_0 , then the curve $x=f(t), y=g(t)$ may or may not be smooth around t_0 .

Ex The curve with parametrization $x=t^3, y=t^6$ is just the parabola $y=x^2$, so it is smooth everywhere although $dx/dt = 3t^2$ and $dy/dt = 6t^5$ both vanish at $t=0$

Horizontal Tangent for Parametric Equations

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \frac{dy}{dt} = 0 \quad \text{provided that} \quad \frac{dx}{dt} \neq 0$$

Vertical Tangent for parametric Equations

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \frac{dx}{dt} = 0 \quad \text{provided that} \quad \frac{dy}{dt} \neq 0$$

Tangents and Normals to Parametric Curves

If f' and g' are continuous and not both 0 at t_0 , then the parametric equations

$$\begin{aligned} x &= f(t_0) + f'(t_0)(t - t_0) & -\infty < t < \infty \\ y &= g(t_0) + g'(t_0)(t - t_0) \end{aligned}$$

represent the tangent line to the parametric curve $x=f(t)$, $y=g(t)$ at the point $(f(t_0), g(t_0))$. The normal line

there has parametric equations

$$\begin{aligned} x &= f(t_0) + g'(t_0)(t - t_0) & (-\infty < t < \infty) \\ y &= g(t_0) - f'(t_0)(t - t_0) & (-\infty < t < \infty) \end{aligned}$$

Both lines pass through $(f(t_0), g(t_0))$ when $t=t_0$

Ex 7 Find equations of the tangent and normal lines to the parametric curve $x = t^2 - t$, $y = t^2 + t$ at point $t = 2$

At $t = 2 \Rightarrow x = 4 - 2 = 2$ $y = 4 + 2 = 6$

$$\frac{dx}{dt} = 2t - 1 \quad \frac{dy}{dt} = 2t + 1 \quad \left. \frac{dx}{dt} \right|_{t=2} = 3 \quad \left. \frac{dy}{dt} \right|_{t=2} = 5$$

Tangent and normal lines

Tangent $x = 2 + 3(t - 2) = 3t - 4$
 $y = 6 + 5(t - 2) = 5t - 4$

Normal $x = 2 + 5(t - 2) = 5t - 8$
 $y = 6 - 3(t - 2) = -3t + 12$

Ex 7 Find equations of the tangent line to the parametric curve given in $x = t^5 - 4t^3$ $y = t^2$ at $(0, 4)$

Firstly, we will obtain for which values of t we can obtain $(0, 4)$

$$x = t^5 - 4t^3 \Rightarrow x = 0 = t^3(t^2 - 4) = t^3(t - 2)(t + 2) \quad \begin{matrix} t = 0 \\ t = \pm 2 \end{matrix}$$

$$\Rightarrow y = t^2 \quad t = 0 \text{ is not satisfied}$$

$t = \pm 2$ to get $(0, 4)$ point

Therefore, we will have

Classical Tangents

$$\frac{dy}{dx} = \frac{2t}{5t^4 - 12t^2}$$

$$\left. \frac{dy}{dx} \right|_{t=2} = \frac{1}{8}$$

$$\left. \frac{dy}{dx} \right|_{t=-2} = -\frac{1}{8}$$

$$y - 4 = \frac{1}{8}(x - 0)$$

AND

$$y - 4 = -\frac{1}{8}(x - 0)$$

Parametric Version

$$\frac{dx}{dt} = 5t^4 - 12t^2$$

$$\left. \frac{dx}{dt} \right|_{t=\pm 2} = 80 - 48 = 32$$

$$t = 2 \quad \left. \frac{dy}{dt} \right|_{t=2} = 4$$

Parametric equations

$$\begin{aligned} x &= 0 + 32(t - 2) \\ y &= 4 + 4(t - 2) \end{aligned}$$

$$t - 2 = \frac{x}{32}$$

$$y = 4 + \frac{1}{8} \frac{x}{32}$$

$$y = 4 + \frac{x}{8}$$

$$t = -2 \quad \left. \frac{dy}{dt} \right|_{t=-2} = 4$$

Parametric version

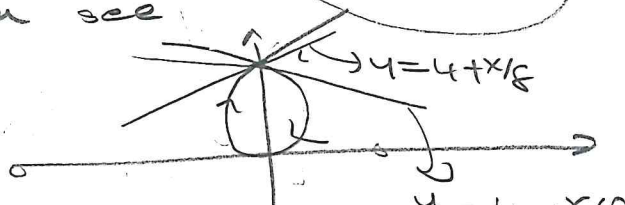
$$\begin{aligned} x &= 0 + 32(t - 2) \\ y &= 4 - 4(t - 2) \end{aligned}$$

$$t - 2 = \frac{x}{32} \Rightarrow y = 4 - \frac{1}{8} \frac{x}{32}$$

$$y = 4 - \frac{x}{8}$$

$\rightarrow$ If we draw the line we can see

parametric curves



Ex Determine the xy-coordinates of the points where the following parametric equations will have horizontal and vertical tangents.

$$x = t^3 - 3t \quad y = 3t^2 - 9$$

Horizontal Tangent

• $\frac{dy}{dt} = 6t = 0$ for $t=0$ So we have horizontal tangent at $(x,y) = (t^3 - 3t, 3t^2 - 9)|_{t=0} = (0, -9)$

$\frac{dx}{dt} = 3t^2 - 3 \neq 0$

Tangent lines Horizontal

$$x = 0 + (-3)t \quad \left. \frac{dx}{dt} \right|_{t=0} = 3t^2 - 3|_{t=0} = -3$$

$$y = -9 + 0 \cdot t \quad y = -9$$

• Vertical Tangent

$\frac{dx}{dt} = 3t^2 - 3 = 0 \quad t = \pm 1$ we have vertical tangent line at $(x,y) = (t^3 - 3t, 3t^2 - 9)$

$\frac{dy}{dt} = 6t \neq 0$

$(x,y)|_{t=1} = (1-3, 3-9) = (-2, -6)$

$(x,y)|_{t=-1} = (-1+3, 3-9) = (2, -6)$

Tangent lines

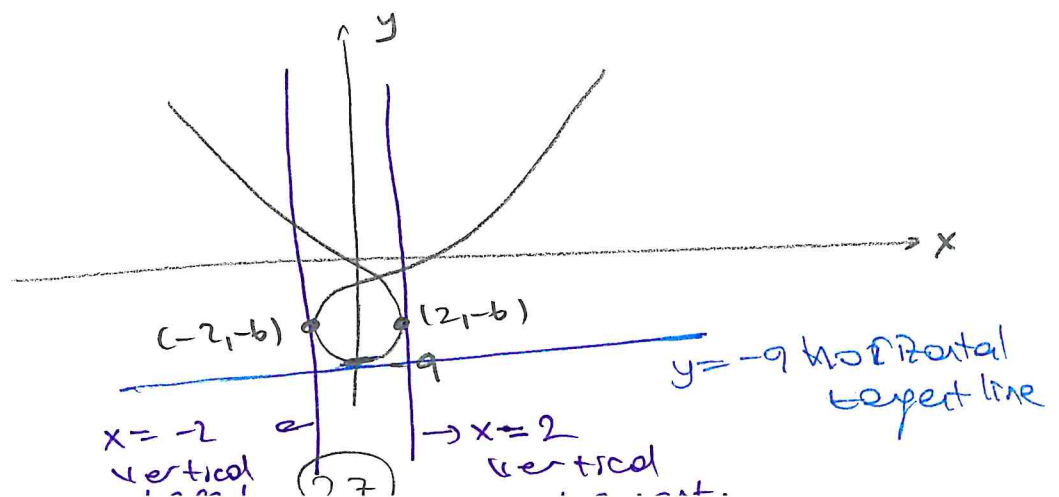
$\left. \frac{dx}{dt} \right|_{t=\pm 1} = 0$

$\left. \frac{dy}{dt} \right|_{t=1} = 6 \Rightarrow x = -2$

$y = -6 + 6(t-1)$

$\left. \frac{dy}{dt} \right|_{t=-1} = -6 \Rightarrow x = 2$

$y = -6 - 6(t+1)$



Concavity of a parametric curve and second derivative

The concavity of a parametric curve can be determined by using the second derivatives of the parametric equations. The procedure is

Just to calculate d^2y/dx^2 using the Chain Rule

$$x = f(t) \quad y = g(t)$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \frac{g'(t)}{f'(t)} = \frac{d}{dt} \left(\frac{g'(t)}{f'(t)} \right) \cdot \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{f'(t)g''(t) - g'(t)f''(t)}{(f'(t))^2} \cdot \frac{1}{f'(t)}$$

→ On an interval where $f'(t) \neq 0$, the parametric curve $x = f(t)$
 $y = g(t)$ has concavity determined by

$$\frac{d^2y}{dx^2} = \frac{f'(t)g''(t) - g'(t)f''(t)}{(f'(t))^3}$$

Ex Use slope and concavity information to sketch the graph of parametric curve

$$x = f(t) = t^3 - 3t \quad y = g(t) = t^2 \quad (-2 \leq t \leq 2)$$

$$\text{we have } f'(t) = 3t^2 - 3 = 3(t-1)(t+1) \quad g'(t) = 2t$$

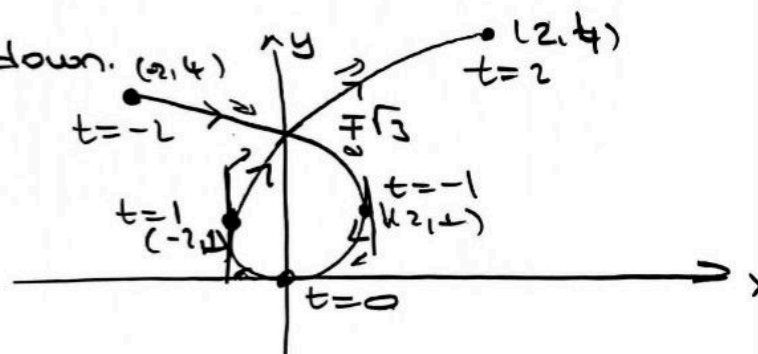
t	-2	-1	0	1	2
f'(t)	+	0	-	0	+
g'(t)	-	-	0	+	+
x	→	•	←	←	→
y	↓	↓	↓	↑	↑
Curve					

x stops only y (2x)
 y stops only
 x stops only

$$\frac{d^2y}{dx^2} = \frac{f'(t)f''(t) - f''(t)f'(t)}{(f'(t))^3} = \frac{3(t^2-1)(2) - 2t \cdot (6t)}{[3(t^2-1)]^3} = -\frac{2}{9} \frac{t^2+1}{(t^2-1)^3}$$

Concave up if $t^2-1 < 0$ $-1 < t < 1$ concave up

Otherwise it is concave down. $(2, 4)$ $t=2$



(SO DIFFICULT FOR STUDENTS)

Ex 7 Determine the values of t for which the parametric curve given by the following set of parametric equations is concave up and concave down.

$$x = 1 - t^2 \quad y = t^7 + t^5$$

For concavity, we deal with $\frac{d^2y}{dx^2}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{7t^6 + 5t^4}{-2t} \quad t \neq 0 = -\frac{1}{2} (7t^5 + 5t^3)$$

$$\frac{dy}{dx} = 0 \Rightarrow 7t^5 + 5t^3 = t^3(7t^2 + 5) = 0 \quad t = 0$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx} = -\frac{1}{2} (35t^4 + 15t^2) \cdot \frac{1}{-2t}$$

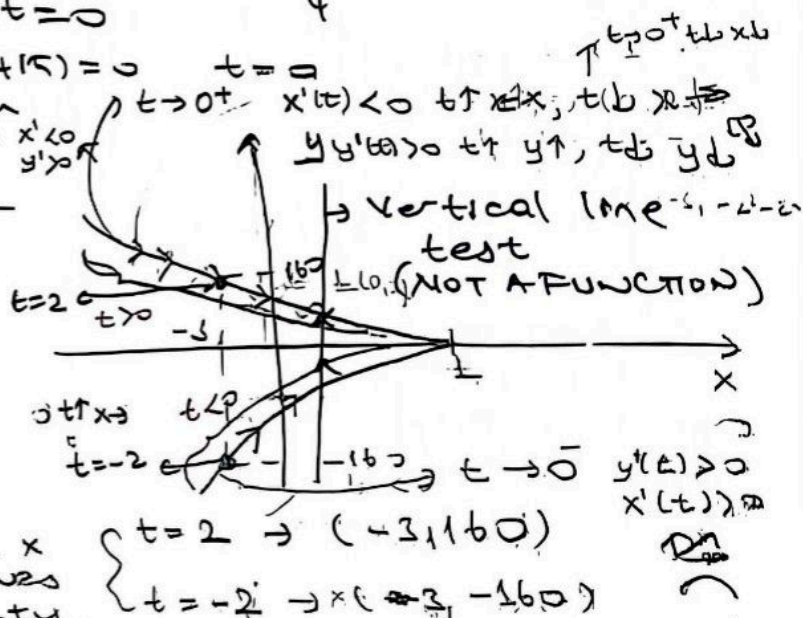
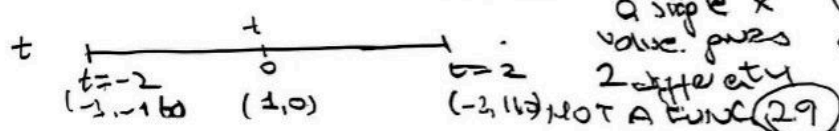
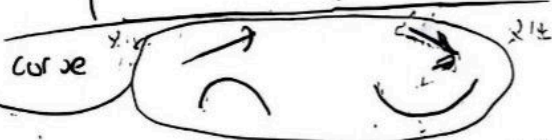
$$= \frac{1}{4} (35t^3 + 15t)$$

$$\frac{d^2y}{dx^2} = 0 \Rightarrow 35t^3 + 15t = 0$$

$$t(35t^2 + 15) = 0 \quad t = 0$$

NOTE: This arrow shows ratio between x, y but for $t > 0$ $x' < 0$ $y' > 0$

t	$t < 0$	$t = 0$	$t > 0$
$\frac{dy}{dx}$	+	undef	-
$\frac{d^2y}{dx^2}$	-	undef	+



$$\begin{cases} t = 2 \rightarrow (-3, 160) \\ t = -2 \rightarrow (-3, -160) \end{cases}$$

ARC LENGTHS AND AREAS OF PARAMETRIC CURVES

→ Let C be a smooth parametric curve with equations

$$x=f(t) \quad y=g(t) \quad a \leq t \leq b$$

(we assume $f'(t)$ and $g'(t)$ are continuous on the interval $[a,b]$ and are never both zero)

→ The length of the curve C is given by

$$S = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Back to our previous information about arc length

length of a function $y=h(x)$ on the interval $[a,b]$

$$L = \int_a^b \sqrt{1 + h'(x)^2} dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx =$$

↑
take dx into square

$$L = \int_a^b \sqrt{\frac{(dx)^2 + (dy)^2}{(dx)^2}} = \int_a^b \sqrt{(dx)^2 + (dy)^2} = \int_a^b ds$$

↓
 $ds = \sqrt{(dx)^2 + (dy)^2}$

For $x=f(t)$, $y=g(t)$ $a \leq t \leq b$ we have

$$L = \int_a^b \sqrt{(dx)^2 + (dy)^2} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Ex 7 Find the length of the parametric curve

$$x=e^t \cos t, \quad y=e^t \sin t \quad (0 \leq t \leq 2)$$

Solution We have $\frac{dx}{dt} = e^t(\cos t - \sin t)$ $\frac{dy}{dt} = e^t(\sin t + \cos t)$

$$\begin{aligned} \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= e^{2t}(\cos t - \sin t)^2 + e^{2t}(\sin t + \cos t)^2 \\ &= e^{2t}(\cos^2 t - 2\cos t \sin t + \sin^2 t) \\ &\quad + e^{2t}(\sin^2 t + 2\cos t \sin t + \cos^2 t) = 2e^{2t} \end{aligned}$$

The length of the curve is, therefore,

$$S = \int_0^2 \sqrt{2e^{2t}} dt = \sqrt{2} \int_0^2 e^t dt = \sqrt{2}(e^2 - 1)$$

(37)

ARC LENGTHS AND AREAS OF PARAMETRIC CURVES

→ If the curve $x=f(t)$ $y=g(t)$ $a \leq t \leq b$

is rotated about the x -axis, the area S of the surface so generated is given by

$$S = 2\pi \int_a^b \underbrace{|g(t)|}_{|y|} \underbrace{\sqrt{(f'(t))^2 + (g'(t))^2}}_{ds} dt$$

→ If the rotation is about the y -axis then the area is

$$S = 2\pi \int_a^b \underbrace{|f(t)|}_{|x|} \underbrace{\sqrt{(f'(t))^2 + (g'(t))^2}}_{ds} dt$$

Previous information

$$y=f(x) \quad x \in [a, b]$$

Rotated x axis $S = 2\pi \int_a^b \underbrace{|f(x)|}_{|y|} \underbrace{\sqrt{1+(f'(x))^2}}_{ds} dx$

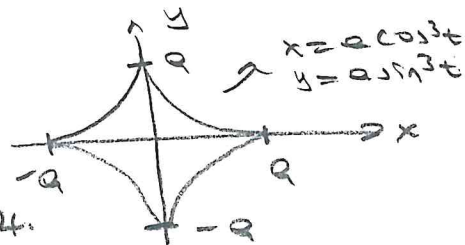
Rotated y axis $S = 2\pi \int_a^b \underbrace{|x|}_{|x|} \underbrace{\sqrt{1+(f'(x))^2}}_{ds} dx$

$|y| = \begin{cases} y & y > 0 \\ -y & y < 0 \end{cases}$
 $|x| = \begin{cases} x & x > 0 \\ -x & x < 0 \end{cases}$

Ex Consider the astroid curve $x=a\cos^3 t$ $y=a\sin^3 t$ ($a>0$)

① Arc length for $0 \leq t \leq 2\pi$

Let's find only the arc length of the first quadrant then multiply it by 4.



$$\frac{dx}{dt} = 3a\cos^2 t (-\sin t) \quad \frac{dy}{dt} = 3a\sin^2 t (\cos t)$$

$$\begin{aligned} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} &= \sqrt{9a^2 \cos^4 t \sin^2 t + 9a^2 \sin^4 t \cos^2 t} \\ &= \sqrt{9a^2 \cos^2 t \sin^2 t (\cos^2 t + \sin^2 t)} \\ &= 3a |\sin t \cos t| \quad t \in [0, \pi/2] \quad \cos t > 0 \quad \sin t > 0 \\ &= 3a \sin t \cos t \quad (31) \end{aligned}$$

$$L = 4 \int_0^{\pi/2} 3a \cos t \sin t dt = 4 \left[-\frac{(\cos t)^2}{2} \right]_0^{\pi/2} \cdot 3a = 6a$$

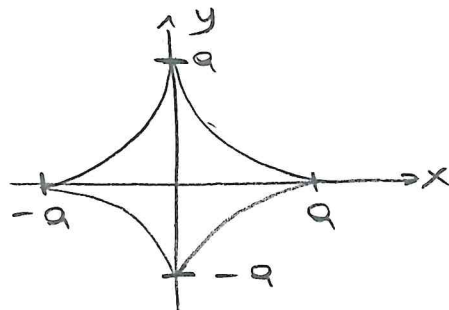
$$\cos t = u$$

$$-\sin t dt = du$$

$$\int \underbrace{\cos t}_{u} \underbrace{\sin t dt}_{-du} = -\frac{u^2}{2} + C$$

- ② Find the area of the surface of revolution obtained by rotating the astroid curve $x = a \cos^3 t$ $y = a \sin^3 t$ ($a > 0$) about the x axis

$$S = 2\pi \int_a^b |y| \sqrt{(f'(t))^2 + (g'(t))^2} dt$$



$$S = 2 \times 2\pi \int_0^{\pi/2} a \sin^3 t \cdot 3a \cos t \sin t dt$$

$$u = \sin t$$

$$du = \cos t dt$$

$$= 4\pi \int_0^{\pi/2} 3a^2 \underbrace{\sin^4 t}_{u^4} \underbrace{\cos t dt}_{du}$$

$$= 12\pi a^2 \frac{\sin^5 t}{5} \Big|_0^{\pi/2} = \frac{12\pi a^2}{5} \text{ square units.}$$

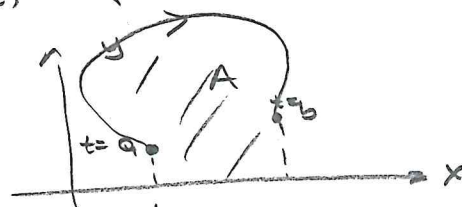
The curve is symmetric about both coordinates. The entire surface will be generated by rotating the upper half of the curve; in fact, we need only rotate first quadrant part and multiply by 2.

Areas Bounded By Parametric Equations

→ Consider the parametric curve C with equations $x = f(t)$, $y = g(t)$ ($a \leq t \leq b$) where f is differentiable and g is continuous on $[a, b]$

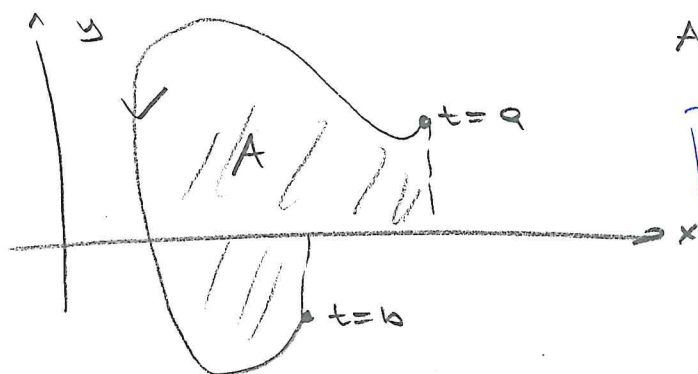
→ If C is a non-self intersecting closed curve, then the area of the region bounded by C is given by

• $A = \int_a^b g(t) f'(t) dt$ If C is traversed clockwise as t increases



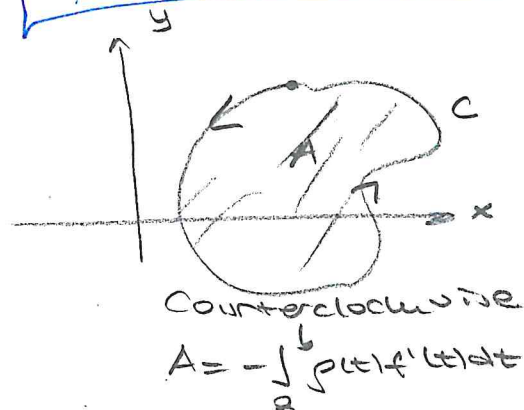
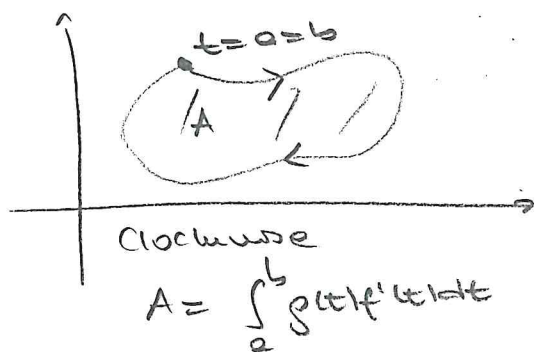
$$A = - \int_a^b g(t) f'(t) dt$$

• $A = - \int_a^b g(t) f'(t) dt$ if C is traversed counterclockwise



$$A = - \int_a^b g(t) f'(t) dt$$

If $f'(t) > 0, g(t) \geq 0, t \in [a, b]$ $A = - \int_a^b g(t) f'(t) dt$
 If $f'(t) \leq 0, g(t) \geq 0, t \in [a, b]$ $A = - \int_a^b g(t) f'(t) dt$
 If $f'(t) \leq 0, g(t) \leq 0, t \in [a, b]$ $A = \int_a^b g(t) f'(t) dt$
 If $f'(t) > 0, g(t) \leq 0, t \in [a, b]$ $A = \int_a^b g(t) f'(t) dt$



Areas bounded by closed parametric curves

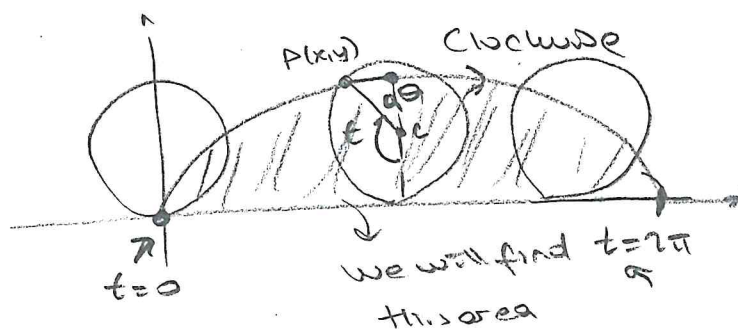
Ex Find the area bounded by the ellipse $x = a \cos s, y = b \sin s$
 $0 \leq s \leq 2\pi$

The ellipse is traversed counterclockwise

$$A = - \int_0^{2\pi} b \cos s (-a \sin s) ds = \frac{ab}{2} \int_0^{2\pi} (1 - \cos 2s) ds$$

$$= \frac{ab}{2} s \Big|_0^{2\pi} - \frac{ab}{4} \sin 2s \Big|_0^{2\pi} = \pi ab \text{ square units}$$

Ex Find the area above the x-axis and under one arch of the cycloid $x = at - a \sin t, y = a - a \cos t$



$$y = a - a \cos t$$

$$y = a(1 - \cos t) \geq 0$$

$$\frac{dx}{dt} = a(1 - \cos t) \geq 0$$

$$A = \int_0^{2\pi} a^2 (1 - \cos t)^2 dt$$

$$= a^2 \int_0^{2\pi} \left(1 - 2\cos t + \frac{1 + \cos 2t}{2} \right) dt$$

$$= a^2 \left(t - 2\sin t + \frac{t}{2} + \frac{\sin 2t}{4} \right) \Big|_0^{2\pi}$$

$$= 3\pi a^2 \text{ square units}$$

Ex. Determine the area under the curve given by the following parametric equations $0 \leq \theta \leq 2\pi$ (counterclockwise)

$$x = 6(\theta - \sin \theta) = f(\theta)$$

$$f'(\theta) = 6(1 - \cos \theta) \geq 0$$

$$y = 6(1 - \cos \theta) = g(\theta) \geq 0$$

$$A = + \int_0^{2\pi} 36(1 - \cos \theta)^2 d\theta = 36 \int_0^{2\pi} (1 - 2\cos \theta + \underbrace{\cos^2 \theta}_{\frac{\cos 2\theta + 1}{2}}) d\theta$$

$$= 36 \int_0^{2\pi} \left(\frac{3}{2} - 2\cos \theta + \frac{\cos 2\theta}{2} \right) d\theta$$

$$= 36 \left(\frac{3}{2}\theta - 2\sin \theta + \frac{\sin 2\theta}{2} \right) \Big|_0^{2\pi} = 108\pi$$

Explanation Page 24

→ Chain Rule Let $x = f(t)$ $y = g(t)$. Substitute them in the equation $y = F(x)$ we get $g(t) = F(f(t))$

— If g, F and f are differentiable, then Chain Rule gives

$$g'(t) = F'(f(t)) \cdot f'(t) = F'(x) \cdot f'(t)$$

— If $f'(t) \neq 0$, we can solve $F'(x) = \frac{g'(t)}{f'(t)}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \text{if} \quad \frac{dx}{dt} \neq 0$$

— Note that we can write $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$ if y is a function of t and t is a function of x But this is not always the case.

— Hence, it is not always correct to write $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$

For instance $x = t^2$ $t = \sqrt{x}$ t is not a function of x

